

$$a_n = \frac{27^n (n!)^3}{(3n)!}$$

1) Find $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$.

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1$$

2) Based on your answer, what does the ratio test say about $S = \sum_{n=1}^{\infty} a_n$?

Choose 1 answer:



INCORRECT

The series converges.



INCORRECT

The series diverges.



CORRECT (SELECTED)

The test is inconclusive.

1 / 2

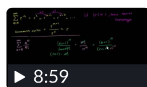
$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1$$

[Show entire process.](#) ✓

2 / 2

Since the limit is equal to 1, the ratio test is inconclusive.

Related content



Ratio test

8:59

[Report a problem](#)